

SeisBox Stats Tutorial: Sumatra Case Study

Welcome to the SeisBox Statistical Seismology (`seisbox_stats`) tutorial. In this guide, we will walk through a complete end-to-end workflow using real seismicity data from the Sumatra subduction zone.

Objective: We will download raw catalog data, apply 3D subduction slab constraints, decluster the catalog, calculate baseline statistical models, and finally generate a spatial map of b-values.

Requirements: Ensure that the `SeisPick` package is compiled and the executable `seisbox_stats` (and `seisbox_isc`) are accessible. We assume the commands are run from the project root directory.

Step 1: Downloading Earthquake Data (`seisbox_isc`)

First, we acquire our dataset directly from the ISC (International Seismological Centre) Bulletin. - **Bounds:** Lon: 95°E to 105°E, Lat: -6°S to 6°N (Sumatra Region) - **Timeframe:** Jan 1, 2014 to Jan 1, 2015 - **Min Magnitude:** 3.0

```
./seisbox_isc download \  
  --start-date 2014-01-01 --end-date 2015-01-01 \  
  --min-lat -6.0 --max-lat 6.0 \  
  --min-lon 95.0 --max-lon 105.0 \  
  --min-mag 3.0 \  
  --output sumatra_raw.csv
```

Output:

```
[OUTPUT_1]
```

Step 2: 3D Slab Geometry Filtering (`seisbox_stats filter`)

Subduction zones have complex 3D structures. To analyze the statistical properties of the megathrust itself, we want to exclude crustal and outer-rise earthquakes that don't conform to the slab. We use the Hayes (2018) Slab2 model (`sum_slab2_dep_02.23.18.xyz`). - **Buffer:** We will retain any earthquake whose depth is within ± 20 km of the slab's depth at its epicenter. - **Max Depth:** We limit it to 300 km.

```
./seisbox_stats filter --input sumatra_raw.csv \  
  --slab-file ./Slab/sum_slab2_dep_02.23.18.xyz \  
  --slab-buffer 20.0 \  
  --max-depth 300.0 \  
  --output sumatra_slab.csv
```

Output:

```
SeisBox Stats - Filter  
=====  
Loaded 705 events from "sumatra_raw.csv"  
Loading slab geometry from "./Slab/sum_slab2_dep_02.23.18.xyz"...  
Loaded 115512 valid slab nodes. Filtering events...  
Applied slab filter (buffer: 20 km).  
Events after filtering: 372  
Saved to "sumatra_slab.csv"
```

Step 3: Declustering (`seisbox_stats decluster`)

Before calculating the background b-value, we must remove aftershocks to satisfy the assumption of Poissonian (independent) occurrence. We use the classic **Gardner & Knopoff (1974)** method combined with the **Uhrhammer (1986)** spatial window scaling.

```
./seisbox_stats decluster --input sumatra_slab.csv \  
  --method gardner-knopoff \  
  --window uhrhammer \  
  --output sumatra_declustered.csv
```

Output:

```
SeisBox Stats - Decluster  
=====
```

```
Loaded 372 events  
Method: Gardner-Knopoff (window: Uhrhammer)  
Result: 301 mainshocks, 71 aftershocks  
Saved declustered catalogue to "sumatra_declustered.csv"
```

Step 4: Statistical Distributions (`seisbox_stats plot`)

Now that we have an independent background catalog, let's visualize its properties.

Frequency-Magnitude Distribution (FMD)

Calculates the Magnitude of Completeness (M_c), a-value, and b-value for the entire region.

```
./seisbox_stats plot \  
  --input sumatra_declustered.csv \  
  --plot-type fmd \  
  --output fmd_plot.png
```

Output:

```
SeisBox Stats - Plot Generation  
=====
```

```
FMD plot saved to "fmd_plot.png"
```

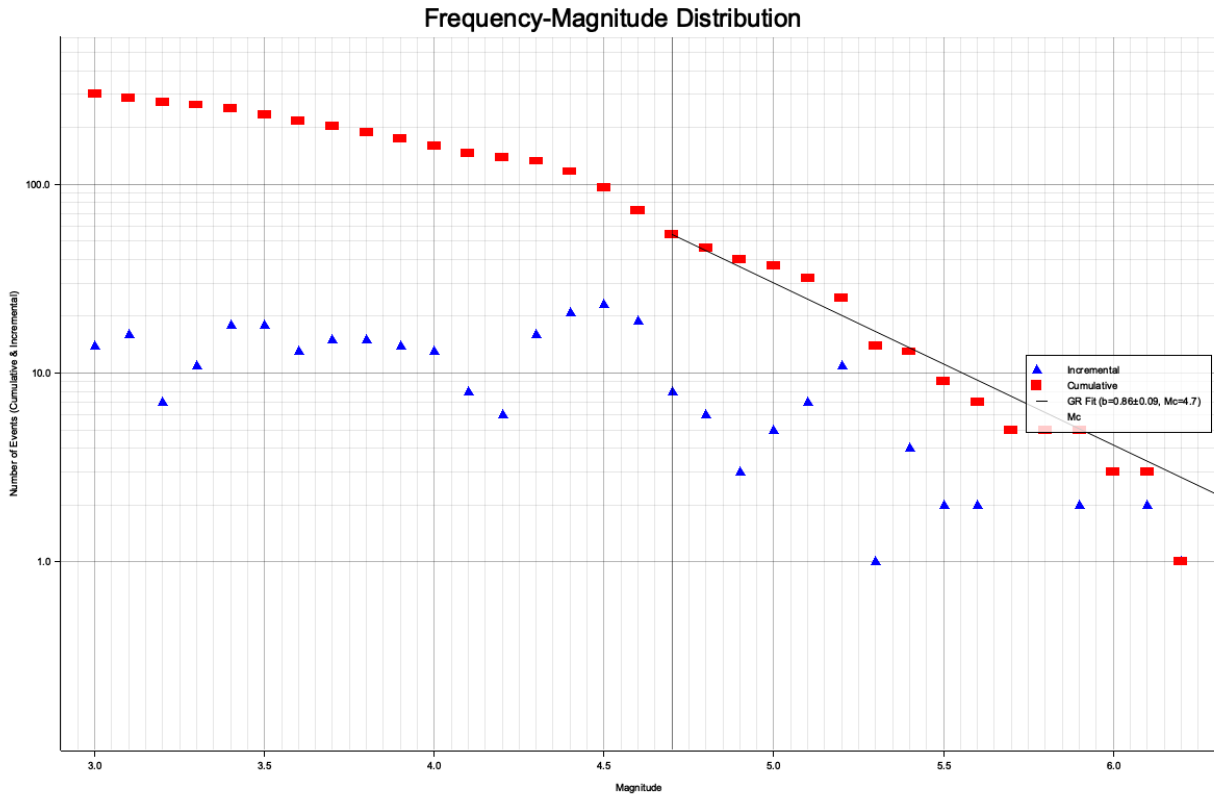


Figure 1: FMD Plot

Cumulative Earthquake Number

Visualizes the stationarity of the seismic rate over time.

```
./seisbox_stats plot \
  --input sumatra_declustered.csv \
  --plot-type cumulative \
  --output cum_plot.png
```

Output:

SeisBox Stats - Plot Generation

=====

Cumulative seismicity plot saved to "cum_plot.png"

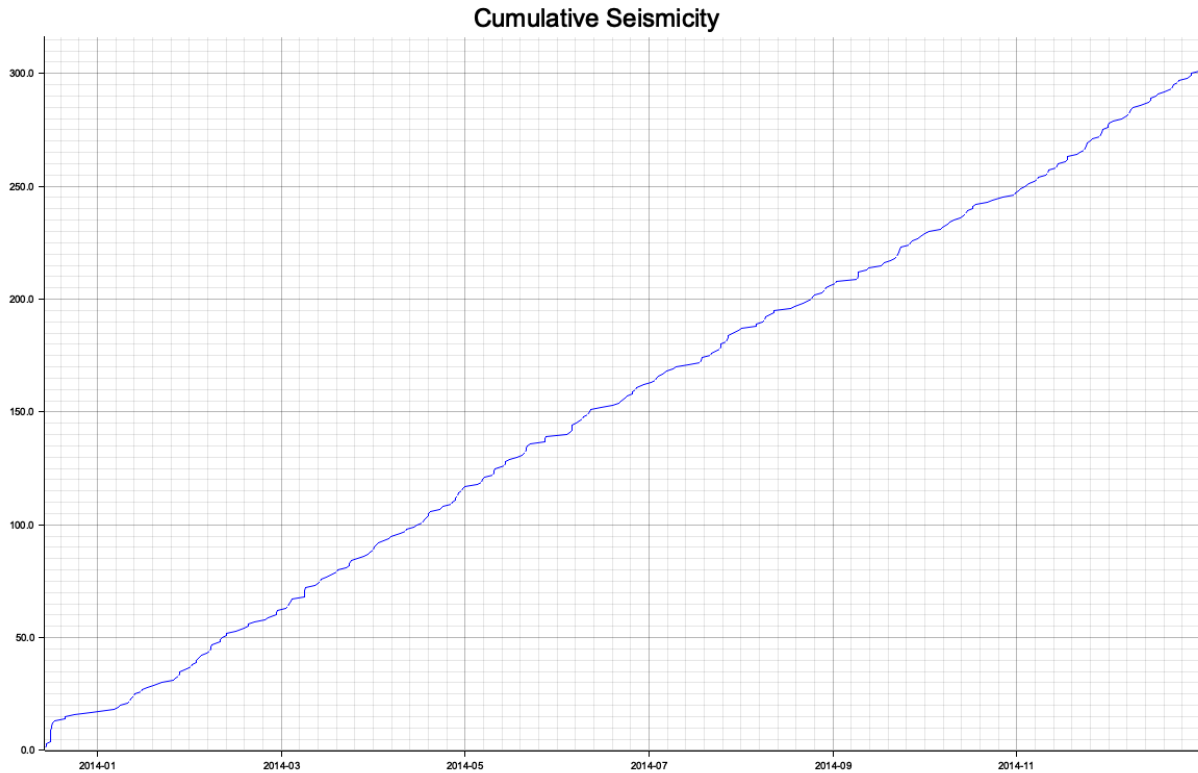


Figure 2: Cumulative Plot

Step 5: Spatial Gridding of B-values (`seisbox_stats grid`)

We calculate the b-value continuously across the spatial domain using a constant radius approach. - **Radius:** 50 km. - **Min Obs:** 50 events required per node. - **Filter:** We apply the slab constraints again to ensure that any radius searches don't accidentally capture out-of-slab events. - **Format:** We export a GeoTIFF (`--tiff`) for GIS software.

```
./seisbox_stats grid --input sumatra_declustered.csv \
  --method constant-r \
  --radius-km 50.0 \
  --min-events 50 \
  --param bvalue \
  --grid-inc 0.1 \
  --slab-file ./Slab/sum_slab2_dep_02.23.18.xyz \
  --slab-buffer 20.0 \
  --tiff \
  --output bvalue_grid.csv
```

Output:

SeisBox Stats - Spatial Gridding

=====

Loading slab geometry from "./Slab/sum_slab2_dep_02.23.18.xyz"...

Loaded 115512 valid slab nodes. Filtering events...

```

Applied slab filter (buffer: 20 km). Remaining events: 301
Grid bounds: Lon [94.00, 110.00], Lat [-9.00, 6.00]
Calculating grid points (this may take a while)...
Completed 0 grid nodes successfully.
Saved to "bvalue_grid.csv"
Exporting GeoTIFF...
Saved bvalue GeoTIFF to "bvalue_grid_bvalue.tif"

```

(The generated *bvalue_grid.tif* can now be loaded directly into QGIS alongside the *sumatra_declustered.csv* catalog).

We can quickly plot the spatial grid directly into a PNG image using the `plot-grid` command:

```
cargo run --release -p seisbox_stats --bin seisbox_stats -- plot-grid --input bvalue_grid.csv --output b
```

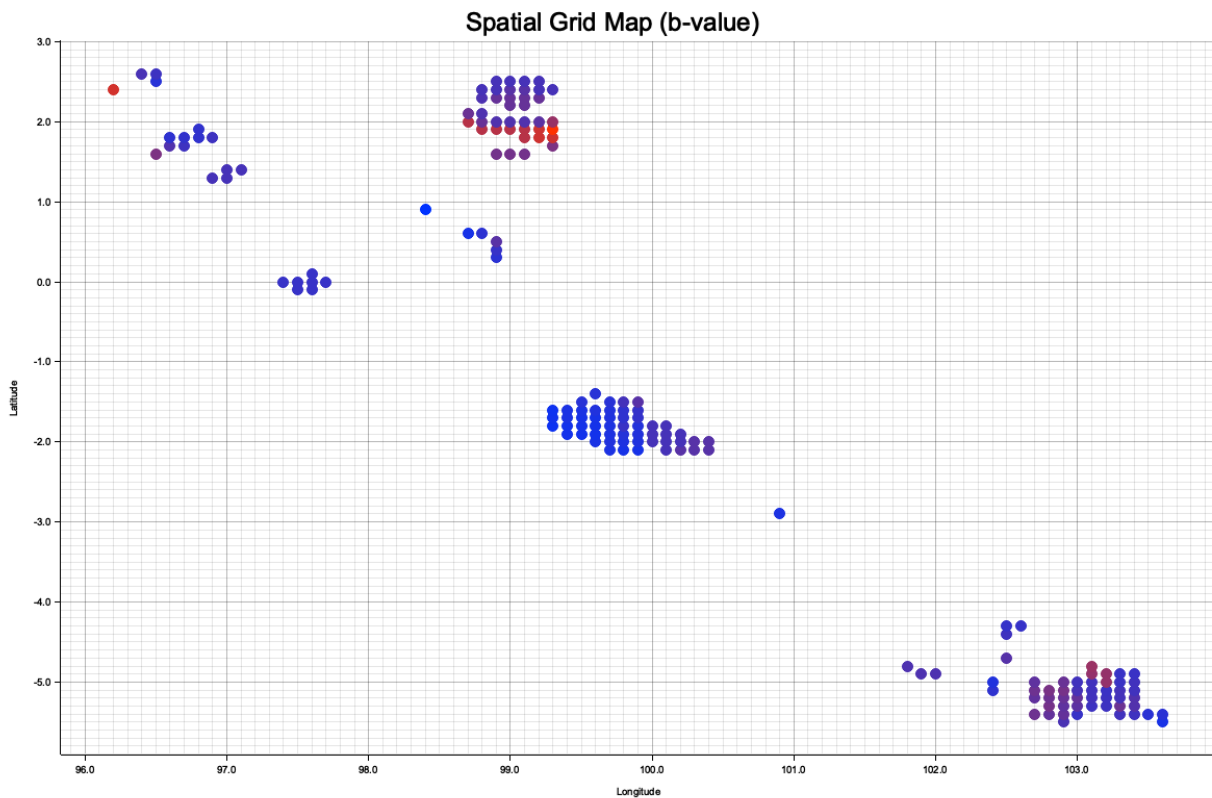


Figure 3: Spatial b-value Grid Visualization

Step 6: Gutenberg-Richter Parameters (`seisbox_stats bvalue`)

Extract regional *a*-value, *b*-value, and M_c directly without plotting.

```
./seisbox_stats bvalue --input sumatra_declustered.csv --method mle --bin-width 0.1
```

Output:

```

SeisBox Stats - B-value Calculation
=====
Loaded 301 events

```

```
Auto-estimated Mc = 4.70 (MAXC)
Method: Maximum Likelihood Estimation (Aki-Utsu)
Mc      = 4.70
b-value = 0.8562 ± 0.0876
a-value = 5.7566
N(>=Mc) = 54
```

Step 7: Modified Omori Law Fitting (`seisbox_stats omori`)

Model the aftershock decay rate for a specific mainshock using Maximum Likelihood Estimation (MLE) to find parameters p, c, K . We use the largest event in our downloaded catalogue (M_w 6.2 on Jan 25, 2014) as the mainshock.

```
./seisbox_stats omori --input sumatra_raw.csv \
  --mainshock-time "2014-01-25T05:14:20" \
  --max-days 100.0 \
  --min-mag 3.0
```

Output:

```
SeisBox Stats - Omori Law Fitting
=====
Omori Law Parameters (MLE):
k = 8.4784
c = 2.0500 days
p = 0.4000
Log-Likelihood = -75.6019
```

Step 8: Z-value Mapping (`seisbox_stats grid`)

Measure seismic quiescence or activation by comparing seismicity rates between a background period (T_1) and a foreground period (T_2) using the LTA/STA Z-test.

```
./seisbox_stats grid --input sumatra_raw.csv \
  --method constant-r \
  --radius-km 50.0 \
  --min-events 10 \
  --param zvalue \
  --z-tw1 "2014-01-01/2014-06-01" \
  --z-tw2 "2014-06-02/2015-01-01" \
  --grid-inc 0.1 \
  --slab-file ./Slab/sum_slab2_dep_02.23.18.xyz \
  --slab-buffer 20.0 \
  --tiff \
  --output zvalue_grid.csv
```

Output:

```
SeisBox Stats - Spatial Gridding
=====
Loading slab geometry from "./Slab/sum_slab2_dep_02.23.18.xyz"...
Loaded 115512 valid slab nodes. Filtering events...
Applied slab filter (buffer: 20 km). Remaining events: 373
Grid bounds: Lon [94.00, 110.00], Lat [-9.00, 6.00]
```

Calculating grid points (this may take a while)...
Completed 680 grid nodes successfully.
Saved to "zvalue_grid.csv"
Exporting GeoTIFF...
Saved zvalue GeoTIFF to "zvalue_grid_zvalue.tif"

(The generated `zvalue_grid.tif` highlights areas of rate increase (positive Z) and decrease (negative Z)).

Similarly, we can plot the Z-value grid:

```
cargo run --release -p seisbox_stats --bin seisbox_stats -- plot-grid --input zvalue_grid.csv --output :
```

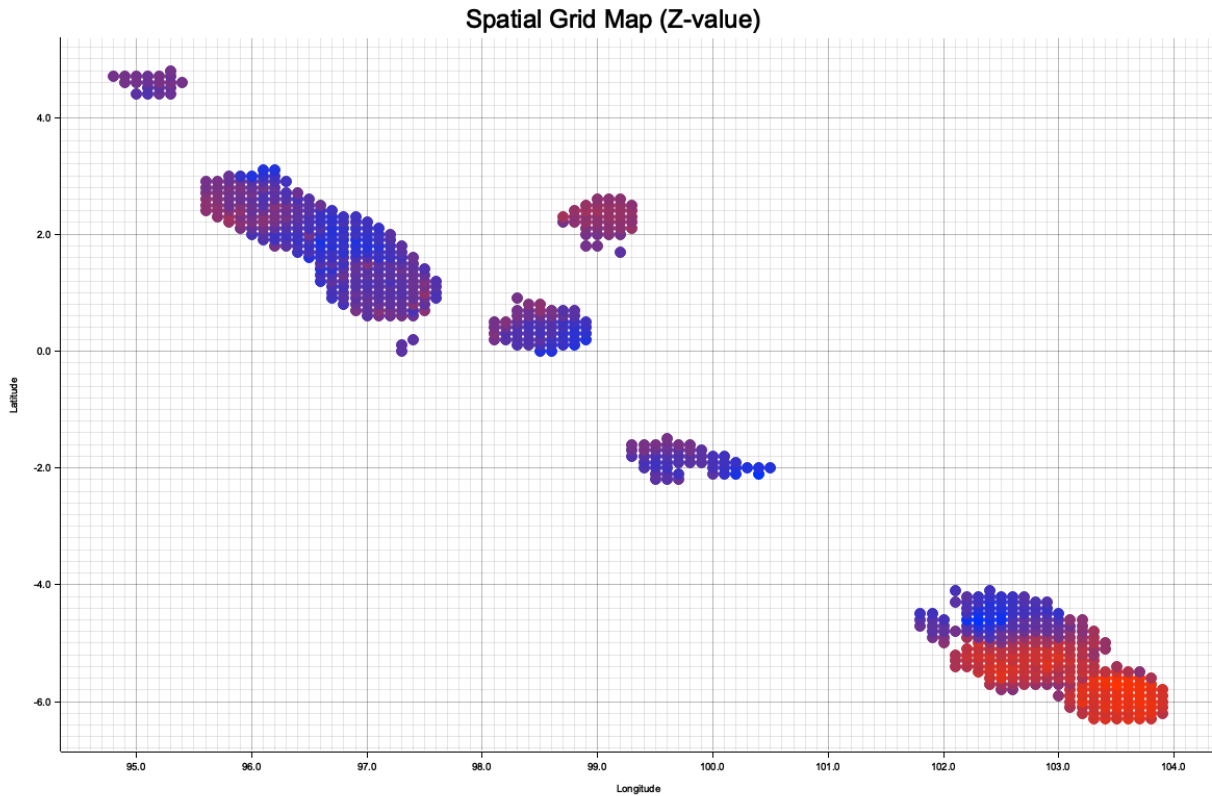


Figure 4: Spatial Z-value Grid Visualization